BEING QUASI-COMPACT IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two Morphisms of Spaces lemmas plus same-chapter items, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Framework lemma for descending properties of morphisms. Stacks Project, Descent on Spaces, Lemma lemma-descending-properties-morphisms. Supplies the criterion structure that the proof verifies.
- Quasi-compactness is \'etale local on the base. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-local. Supplies assumptions (1), (2) and the affine test reduction.
- Surjection from a quasi-compact source. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-surjection-from-quasi-compact. Concludes $Y \times_Z X$ is quasi-compact.

Source reference. Original lemma 041J: sources/stacks-chow/tags.txt, source lines 5294–5294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03KG: sources/stacks-chow/tags.txt, source lines 4676–4676. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 040W: sources/stacks-chow/tags.txt, source lines 5271–5271. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. The property $\mathcal{P}(f) =$"f is quasi-compact" is fpqc local on the base on algebraic spaces over S.*

Proof. We will use Lemma 041J to prove this. Assumptions (1) and (2) of that lemma follow from Morphisms of Spaces, Lemma 03KG. Let $Z' \to Z$ be a surjective flat morphism of affine schemes over S. Let $f : X \to Z$ be a morphism of algebraic spaces, and assume that the base change $f' : Z' \times_Z X \to Z'$ is quasi-compact. We have to show that f is quasi-compact. To see this, using Morphisms of Spaces, Lemma 03KG again, it is enough to show that for every affine scheme Y and

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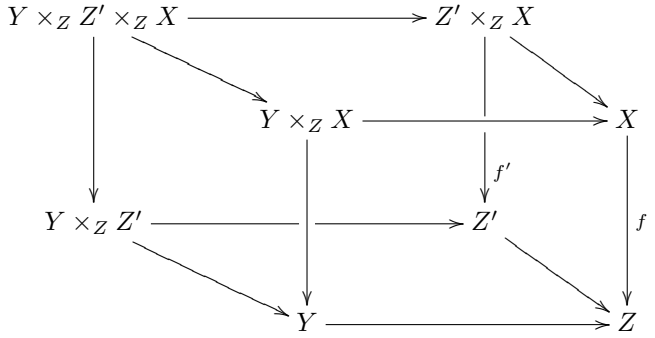
morphism $Y \to Z$ the fibre product $Y \times_Z X$ is quasi-compact. Here is a picture:

(0.1.1)



Note that all squares are cartesian and the bottom square consists of affine schemes. The assumption that f' is quasi-compact combined with the fact that $Y \times_Z Z'$ is affine implies that $Y \times_Z Z' \times_Z X$ is quasi-compact. Since

$$Y \times_Z Z' \times_Z X \longrightarrow Y \times_Z X$$

is surjective as a base change of $Z' \to Z$ we conclude that $Y \times_Z X$ is quasi-compact, see Morphisms of Spaces, Lemma 040W. This finishes the proof. □